IMO 2026 — Problem 2

Solution by Claude Fable 5 (lightweight harness)

2026-07-18

Source: `results/imo-2026/claude-fable-5-lightweight/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC, respectively. Let points K and L be chosen inside triangles BMC and BNC, respectively, such that K lies inside the angle LBA, L lies inside the angle ACK, and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL. Prove that $OM = ON$.

Solution document

Status

solved

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Approaches tried

- **Power-of-a-point reformulation (successful).** Via the median-length formula, $OM = ON \iff \mathrm{pow}(B,\omega) - \mathrm{pow}(C,\omega) = \frac{1}{2}(AB^2 - AC^2)$, where ω is the circumcircle of AKL.
- **Auxiliary points X, Y (successful).** $Y :=$ ray $BK \cap AC$, $X :=$ ray $CL \cap AB$ (crossbar). The three angle hypotheses translate, via directed angles, into three concyclicities: B, X, Y, C on ω_1; C, M, K, X on ω_K; B, N, L, Y on ω_L (with tangency of AB to ω_K at M in the degenerate case $X = M$, and of AC to ω_L at N when $Y = N$).
- **Affine power identity (the key step).** $F(Z) := \mathrm{pow}(Z,\omega) - \mathrm{pow}(Z,\omega_K) - \mathrm{pow}(Z,\omega_L) + \mathrm{pow}(Z,\omega_1)$ is an affine function of Z; it vanishes at the three non-collinear points A, K, L, hence identically. Evaluating at B and C gives $\mathrm{pow}(B,\omega) = \vec{BM} \cdot \vec{BX}$ and $\mathrm{pow}(C,\omega) =$

$\vec{CN}\cdot\vec{CY}$; together with $AB\cdot AX = AC\cdot AY$ (power of A w.r.t. ω_1) and the median formula this yields $OM = ON$.
- **Dead end avoided:** first tried radical-axis arguments with the second intersection points K^*, L^* of line BY with ω_K and of CX with ω_L; it works but requires awkward tangency case analysis. The affine function F replaces all of it.
- Numerical verification: `code/verify.py`, `code/check_signs.py`, `code/verify2.py` confirm every claim (several triangles; generic case; both tangency cases $X = M$ and $Y = N$; all signed-angle sign claims; $F \equiv 0$; $OM = ON$) to machine precision.

Current best

Complete rigorous proof below. Core of the argument: with $X = \text{ray } CL \cap AB$, $Y = \text{ray } BK \cap AC$, the hypotheses force B, X, Y, C, C, M, K, X and B, N, L, Y to be concyclic; then the affine function $\text{pow}(\cdot,\omega) - \text{pow}(\cdot,\omega_K) - \text{pow}(\cdot,\omega_L) + \text{pow}(\cdot,\omega_1)$ vanishes at A, K, L, hence everywhere; evaluating at B, C and using $AB\cdot AX = AC\cdot AY$ and the median-length formula gives $OM = ON$.

Full proof

0. Conventions and notation

We identify the plane with $\mathbb{C}$. For $u, v \in \mathbb{C}$ write

$$\langle u, v\rangle = \text{Re}(\bar{u}v), \qquad u \times v = \text{Im}(\bar{u}v),$$

the usual dot and cross products of plane vectors.

Unsigned angles. For points $P, Q \neq V$, $\angle PVQ \in [0^\circ, 180^\circ]$ is the ordinary (unsigned) angle, i.e. $\angle PVQ = |\arg\frac{Q-V}{P-V}|$ with $\arg \in (-180^\circ, 180^\circ]$.

Signed ray angles. For nonzero u, v let $\measuredangle(u \to v) := \arg\frac{v}{u} \in (-180^\circ, 180^\circ]$. For rays from V through P and through Q put $\measuredangle(VP \to VQ) := \measuredangle((P-V) \to (Q-V))$. Since $\frac{v}{u} = \frac{\bar{u}v}{|u|^2}$ we have:

> **(0.1)** $|\measuredangle(u \to v)| = \angle(u, v)$ (the unsigned angle), and if $u \times v \neq 0$ then the sign of $\measuredangle(u \to v)$ equals the sign of $u \times v$. Moreover $\measuredangle(u \to w) \equiv \measuredangle(u \to v) + \measuredangle(v \to w)$ (mod 360°) and $\measuredangle(v \to u) \equiv -\measuredangle(u \to v)$ (mod 360°) (because arg is additive mod 360° under multiplication and $\arg(u/v) = -\arg(v/u)$ mod 360°).

Directed line angles. For lines ℓ_1, ℓ_2 with direction vectors $d_1, d_2 \neq 0$ set

$$\angle(\ell_1, \ell_2) := \arg\tfrac{d_2}{d_1} \mod 180^\circ.$$

This is well defined: replacing d_i by a nonzero real multiple changes $\arg\frac{d_2}{d_1}$ by 0° or 180°. It is additive mod 180°, and $\angle(\ell_1, \ell_2) \equiv 0$ iff $\ell_1 \parallel \ell_2$ (including $\ell_1 = \ell_2$). If rays r_1, r_2 span the lines ℓ_1, ℓ_2, then $\angle(\ell_1, \ell_2) \equiv \measuredangle(r_1 \to r_2)$ (mod 180°). For points we write $\angle(PQ, RS)$ for the directed angle from line PQ to line RS.

Power of a point. For a circle Γ with centre Z_0 and radius $\rho > 0$ and a point Z, $\text{pow}(Z, \Gamma) := |Z - Z_0|^2 - \rho^2$.

Sides of a line. For a line through P with direction d, points Z_1, Z_2 are *strictly on the same side* iff $d \times (Z_1 - P)$ and $d \times (Z_2 - P)$ are nonzero of equal sign; *strictly opposite sides* iff nonzero of opposite signs.

Interior of a triangle / of an angle. For an affinely independent triple P, Q, R, the (open) interior of triangle PQR is $\{\alpha P + \beta Q + \gamma R : \ \alpha, \beta, \gamma > 0, \ \alpha + \beta + \gamma = 1\}$; "inside triangle PQR" means lying in this set. For $0^\circ < \angle QVR < 180^\circ$, "$P$ lies inside angle QVR" means: P is strictly on the same side of line VQ as R and strictly on the same side of line VR as Q.

Throughout, $b := AC = |C - A|$, $c := AB = |B - A|$, and

$$\theta := \angle KBA = \angle ACL, \qquad \varphi := \angle LBK = \angle LNC, \qquad \psi := \angle LCK = \angle BMK,$$

ω denotes the circumcircle of triangle AKL and O its centre, $R_\omega = |OA|$ its radius. Since the problem speaks of *triangle* AKL and its circumcentre, the points A, K, L are pairwise distinct and not collinear; this is part of the hypothesis.

Reflections, rotations and translations of the plane preserve all hypotheses and the conclusion, so **we may and do assume that ABC is positively oriented**:

$$\delta := (B - A) \times (C - A) > 0. \qquad (0.2)$$

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1. Preliminary lemmas

Lemma 1 (line–circle intersections and power). Let Γ be a circle with centre Z_0, radius $\rho > 0$, and let ℓ be a line. (i) $\ell \cap \Gamma$ has at most two points. (ii) If $M \in \Gamma$, there is exactly one line through M meeting Γ only at M (the **tangent** at M); its direction is $i(M - Z_0)$, i.e. it is the line through M perpendicular to MZ_0. (iii) If $Z \in \ell$ and ℓ meets Γ at U and V — where in the tangent case we set $U = V$ — then

$$\mathrm{pow}(Z, \Gamma) = \langle U - Z, \ V - Z \rangle,$$

the product of signed lengths $ZU \cdot ZV$ along ℓ.

Proof. Parametrize $\ell = \{Z + td : \ t \in \mathbb{R}\}$ with $|d| = 1$, $Z \in \ell$ arbitrary. Then

$$|Z + td - Z_0|^2 - \rho^2 = t^2 + 2t\,\langle d, Z - Z_0 \rangle + \mathrm{pow}(Z, \Gamma), \qquad (1.1)$$

a monic real quadratic in t; it has at most two roots, proving (i). For (ii): take $Z = M \in \Gamma$; then $\mathrm{pow}(M, \Gamma) = 0$ and the roots are $t = 0$ and $t = -2\langle d, M - Z_0 \rangle$; they coincide iff $\langle d, M - Z_0 \rangle = 0$, i.e. iff $d \perp (M - Z_0)$, i.e. iff d is a real multiple of $i(M - Z_0)$; that determines a unique line through M. For (iii): if $U = Z + t_1 d$, $V = Z + t_2 d$ (with $t_1 = t_2$ in the tangent case, when (1.1) has a double root), then by Vieta $t_1 t_2 = \mathrm{pow}(Z, \Gamma)$, and $\langle U - Z, V - Z \rangle = t_1 t_2 \langle d, d \rangle = t_1 t_2$. $\square$

Lemma 2 (directed inscribed-angle criterion). Let P, Q, R be pairwise distinct, non-collinear points and let Γ be the (unique) circle through them. Then for every point $S \notin \{P, Q\}$:

$$S \in \Gamma \iff \angle(SP, \ SQ) \equiv \angle(RP, \ RQ) \pmod{180^\circ}.$$

Proof. Uniqueness of Γ is classical (its centre is the common point of the perpendicular bisectors of PQ and PR, which are not parallel).

Put $w := (Q - R)\overline{(P - R)} \neq 0$. Since $\frac{Q-R}{P-R} = \frac{w}{|P-R|^2}$ and P, Q, R are not collinear, $\arg \frac{Q-R}{P-R} \not\equiv 0 \pmod{180^\circ}$, i.e. $w \notin \mathbb{R}$, so $\operatorname{Im} w \neq 0$.

Define $g : \mathbb{C} \to \mathbb{R}$ by $g(Z) := \operatorname{Im}\left[w\,(P-Z)\overline{(Q-Z)}\right]$. Expanding $(P-Z)\overline{(Q-Z)} = P\bar{Q} - P\bar{Z} - \bar{Q}Z + |Z|^2$ shows

$$g(Z) = (\operatorname{Im} w)\,|Z|^2 + \lambda(Z)$$

with λ a real affine function of $Z \in \mathbb{R}^2$. Hence $\frac{1}{\operatorname{Im} w}\,g(Z) = |Z - Z_1|^2 - \rho_1$ for some $Z_1 \in \mathbb{C}, \rho_1 \in \mathbb{R}$, so the zero set $\{g = 0\}$ is empty, a single point, or a circle. Now

$$g(P) = 0, \qquad g(Q) = 0, \qquad g(R) = \operatorname{Im}\left[(Q-R)\overline{(P-R)}(P-R)\overline{(Q-R)}\right] = \operatorname{Im}\left[|P-R|^2|Q-R|^2\right] = 0.$$

So $\{g = 0\}$ contains three distinct non-collinear points, hence it is a circle through P, Q, R; by uniqueness $\{g = 0\} = \Gamma$.

Finally, fix $S \notin \{P, Q\}$. All four numbers $Q - S$, $P - S$, $Q - R$, $P - R$ are nonzero, and

$$\angle(SP, SQ) \equiv \angle(RP, RQ) \pmod{180^\circ} \iff \arg\left[\tfrac{Q-S}{P-S}\right] \equiv \arg\left[\tfrac{Q-R}{P-R}\right] \pmod{180^\circ} \iff \tfrac{Q-S}{P-S}\cdot\overline{\left(\tfrac{Q-R}{P-R}\right)} \in \mathbb{R}\backslash\{0\}.$$

Multiplying by the positive real $|P-S|^2|P-R|^2$, the last condition is equivalent to $(Q-S)\overline{(P-S)}\cdot\overline{(Q-R)}(P-R) \in \mathbb{R}$, i.e. (taking conjugates) to $w\,(P-S)\overline{(Q-S)} \in \mathbb{R}$, i.e. to $g(S) = 0$, i.e. to $S \in \Gamma$. $\square$

Lemma 3 (directed tangent–chord). Let Γ be a circle through the pairwise distinct points M, K, C, and let t be the tangent to Γ at M. Then (i) $\angle(t,\, MK) \equiv \angle(CM,\, CK) \pmod{180^\circ}$; (ii) if a line ℓ through M satisfies $\angle(\ell,\, MK) \equiv \angle(CM,\, CK) \pmod{180^\circ}$, then $\ell = t$; in particular ℓ meets Γ only at M.

Proof. (i) Maps $Z \mapsto aZ + \beta$ $(a \neq 0)$ preserve directed line angles (all direction vectors get multiplied by a, so ratios of directions are unchanged), map circles to circles and tangent lines to tangent lines (they are bijections preserving the intersection pattern). Hence we may assume Γ is the unit circle centred at 0. Write $m, k, c \in \Gamma$, so $|m| = |k| = |c| = 1$, $\bar{m} = 1/m$ etc. By Lemma 1(ii), t has direction im. Consider

$$z := \frac{(k-m)(m-c)}{m\,(k-c)}.$$

Then

$$\bar{z} = \frac{(\bar{k}-\bar{m})(\bar{m}-\bar{c})}{\bar{m}(\bar{k}-\bar{c})} = \frac{\frac{m-k}{km}\cdot\frac{c-m}{mc}}{\frac{1}{m}\cdot\frac{c-k}{kc}} = \frac{(m-k)(c-m)}{m\,(c-k)} = -z,$$

so z is purely imaginary; also $z \neq 0$. Hence $z/i \in \mathbb{R}\setminus\{0\}$. Now

$$\angle(t, MK) - \angle(CM, CK) \equiv \arg\frac{k-m}{im} - \arg\frac{k-c}{m-c} = \arg\left[\frac{(k-m)(m-c)}{im(k-c)}\right] = \arg\frac{z}{i} \equiv 0 \pmod{180^\circ}.$$

(ii) From (i), $\angle(\ell, MK) \equiv \angle(t, MK)$, hence $\angle(\ell, t) \equiv 0$, so $\ell \parallel t$; both pass through M, so $\ell = t$. $\square$

Lemma 4 (median-length formula). For any points O, P, Q, if $W = \frac{1}{2}(P+Q)$ then

$$4\,|OW|^2 = 2\,|OP|^2 + 2\,|OQ|^2 - |PQ|^2.$$

Proof. With vectors from O: $4|W-O|^2 = |(P-O)+(Q-O)|^2 = |P-O|^2 + 2\langle P-O, Q-O\rangle + |Q-O|^2$ and $|PQ|^2 = |P-O|^2 - 2\langle P-O, Q-O\rangle + |Q-O|^2$. Add. $\square$

Lemma 5 (angle addition inside an angle). Let V, Q, R be points with $0^\circ < \angle QVR < 180^\circ$ and let P lie inside angle QVR. Then $\angle QVP + \angle PVR = \angle QVR$, and $0^\circ < \angle QVP < \angle QVR$.

Proof. Unsigned angles and sides of lines are invariant under translations, rotations and reflections, so we may place $V = 0$ and assume the unit vector along VQ is $q = 1$ and $\gamma := \angle QVR \in (0^\circ, 180^\circ)$ with unit vector $r = (\cos\gamma, \sin\gamma)$ along VR (reflect in the x-axis if needed). Let $p = (\cos\sigma, \sin\sigma)$, $\sigma \in (-180^\circ, 180^\circ]$, be the unit vector along VP. "P on the same side of line VQ (the x-axis) as R" means $\sin\sigma > 0$, so $\sigma \in (0^\circ, 180^\circ)$. "$P$ on the same side of line VR as Q" means $r \times p$ and $r \times q$ have the same sign; $r \times q = -\sin\gamma < 0$ and $r \times p = \sin(\sigma - \gamma)$; since $\sigma - \gamma \in (-180^\circ, 180^\circ)$, the condition $\sin(\sigma - \gamma) < 0$ means $\sigma < \gamma$. So $0 < \sigma < \gamma$, and then $\angle QVP = \sigma$, $\angle PVR = \gamma - \sigma$, $\angle QVR = \gamma$. $\square$

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2. Position facts

Write points as affine combinations $Z = \alpha_Z A + \beta_Z B + \gamma_Z C$ with $\alpha_Z + \beta_Z + \gamma_Z = 1$ (barycentric coordinates w.r.t. ABC; not to be confused with the side lengths b, c). From $Z - A = \beta_Z(B - A) + \gamma_Z(C - A)$, $Z - B = \alpha_Z(A - B) + \gamma_Z(C - B)$, $Z - C = \alpha_Z(A - C) + \beta_Z(B - C)$ and the cyclic identity

$$(B - A) \times (C - A) = (C - B) \times (A - B) = (A - C) \times (B - C) = \delta \qquad (2.1)$$

(direct expansion of all three gives $A \times B + B \times C + C \times A$), we get

$$(B-A) \times (Z-A) = \gamma_Z\,\delta, \qquad (C-B) \times (Z-B) = \alpha_Z\,\delta, \qquad (A-C) \times (Z-C) = \beta_Z\,\delta. \qquad (2.2)$$

In particular, by (0.2): Z is strictly on the same side of line AB as C iff $\gamma_Z > 0$; strictly on the same side of BC as A iff $\alpha_Z > 0$; strictly on the same side of CA as B iff $\beta_Z > 0$. The interior of triangle ABC is exactly $\{\alpha_Z, \beta_Z, \gamma_Z > 0\}$.

(P1) K lies strictly inside triangle ABC and $K \notin$ lines AB, BC, CM. Indeed $K = \alpha B + \mu M + \gamma C$ with $\alpha, \mu, \gamma > 0$, $\alpha + \mu + \gamma = 1$ (B, M, C are affinely independent since M is an interior point of AB and $C \notin AB$). As $M = \frac{1}{2}(A + B)$,

$$K = \tfrac{\mu}{2}A + (\alpha + \tfrac{\mu}{2})B + \gamma C,$$

all coefficients positive, so K is interior to ABC. In coordinates w.r.t. B, M, C: line $BM = AB$ is $\{\gamma = 0\}$, line BC is $\{\mu = 0\}$, line CM is $\{\alpha = 0\}$; the interior has $\alpha, \mu, \gamma > 0$.

(P2) L lies strictly inside triangle ABC and $L \notin$ lines AC, BC, BN. Same argument with $L = \alpha' B + \nu N + \gamma' C$, $N = \frac{1}{2}(A + C)$: $L = \frac{\nu}{2}A + \alpha' B + (\gamma' + \frac{\nu}{2})C$; line $NC = AC$ is $\{\alpha' = 0\}$, line BC is $\{\nu = 0\}$, line BN is $\{\gamma' = 0\}$.

In particular $A \neq K$, $A \neq L$ (A is not interior), $K \notin AB$, $L \notin AC$, so

$$0^\circ < \theta = \angle KBA < 180^\circ, \qquad 0^\circ < \varphi = \angle LNC < 180^\circ, \qquad 0^\circ < \psi = \angle BMK < 180^\circ. \qquad (2.3)$$

(P3) Definition and position of Y and X. *The ray from B through K meets segment AC at a point Y with $Y \neq A, C$; the ray from C through L meets segment AB at a point X with $X \neq A, B$.*

Proof for Y. Let K have barycentric coordinates $(\alpha_K, \beta_K, \gamma_K)$, all positive, by (P1). First, A and C are strictly on opposite sides of line BK (direction $K - B$):

$$(K-B)\times(A-B) = -(A-B)\times(K-B), \qquad (A-B)\times(K-B) = (A-B)\times(K-A) = -(B-A)\times(K-A) = -\gamma_K\,\delta < 0$$

so $(K-B)\times(A-B) = \gamma_K\,\delta > 0$; and by (2.2), $(K-B)\times(C-B) = -(C-B)\times(K-B) = -\alpha_K\,\delta < 0$. (Here $(A-B)\times(K-B) = (A-B)\times(K-A)$ because $(A-B)\times(A-B) = 0$.) Hence segment AC meets line BK in exactly one point Y, and $Y \neq A, C$ (strict signs), $Y \neq B$ (as $B \notin AC$, else $\delta = 0$).

Next, Y lies on the ray from B through K. Write $Y = B + t(K-B)$, $t \neq 0$. On one hand, Y has barycentric coordinates $(1-s, 0, s)$ with $s \in (0,1)$, so by (2.2) $(B-A)\times(Y-A) = s\,\delta > 0$. On the other hand $Y - A = (B-A) + t(K-B)$ and $(B-A)\times(K-B) = (B-A)\times(K-A) = \gamma_K\,\delta$, so $(B-A)\times(Y-A) = t\,\gamma_K\,\delta$. Comparing, $t > 0$.

Proof for X. Symmetric. Let L have barycentric coordinates $(\alpha_L, \beta_L, \gamma_L)$, all positive. Then $(L-C)\times(A-C) = -(A-C)\times(L-C) = -\beta_L\,\delta < 0$ by (2.2), while from $L - C = \alpha_L(A-C) + \beta_L(B-C)$ and (2.1), $(L-C)\times(B-C) = \alpha_L\,(A-C)\times(B-C) = \alpha_L\,\delta > 0$. So A, B are strictly on opposite sides of line CL, and segment AB meets line CL in a single point $X \neq A, B$. Writing $X = C + t(L-C)$, and noting that X has barycentric coordinates $(1-s, s, 0)$ with $s \in (0,1)$, (2.2) gives $(A-C)\times(X-C) = \beta_X\,\delta = s\,\delta > 0$, while $(A-C)\times(X-C) = t\,(A-C)\times(L-C) = t\,\beta_L\,\delta$; hence $t > 0$. □

Introduce the signed coordinates along the two sides:

$$x := AX \in (0, c), \qquad y := AY \in (0, b),$$

so that on the line AB (coordinatized from A towards B) the points A, X, M, B sit at $0, x, \frac{c}{2}, c$, and on line AC the points A, Y, N, C sit at $0, y, \frac{b}{2}, b$.

Note for later use: since Y lies on ray BK and $Y \neq B$, **line** BY = **line** BK; since X lies on ray CL and $X \neq C$, **line** CX = **line** CL. Also $X \neq Y$ (lines AB and AC meet only at A, and $X \neq A$).

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3. Translating the hypotheses into directed angles

By (P1)–(P2), K and L are interior to ABC; let their barycentric coordinates be positive. We use (2.2) and (0.1) repeatedly. Below, "Z is on the C-side of AB" abbreviates "strictly on the same side of line AB as C", etc.

(S1) If Z is on the C-side of AB (i.e. $\gamma_Z > 0$), then $\measuredangle(BA \to BZ) = -\angle ABZ$ and $\measuredangle(MB \to MZ) = +\angle BMZ$. *Proof.* $(A-B)\times(Z-B) = -(B-A)\times(Z-B) = -(B-A)\times(Z-A) = -\gamma_Z\,\delta < 0$, so by (0.1) the signed angle $\measuredangle(BA \to BZ)$ is negative, of absolute value $\angle ABZ$. Also $B - M = \frac{1}{2}(B-A)$ and $(B-A)\times(Z-M) = (B-A)\times(Z-A) - (B-A)\times(M-A) = \gamma_Z\,\delta - 0 > 0$, so $\measuredangle(MB \to MZ) = +\angle BMZ$. □

(S2) If Z is on the B-side of AC (i.e. $\beta_Z > 0$), then $\measuredangle(CA \to CZ) = +\angle ACZ$ and $\measuredangle(NC \to NZ) = -\angle CNZ$. *Proof.* By (2.2), $(A-C)\times(Z-C) = \beta_Z\,\delta > 0$, so by (0.1) $\measuredangle(CA \to CZ) = +\angle ACZ$. For the second claim: $C - N = \frac{1}{2}(C-A)$, and since $N - A$ is parallel to $C - A$,

$$(C-N)\times(Z-N) = \tfrac{1}{2}\,(C-A)\times(Z-N) = \tfrac{1}{2}\,(C-A)\times(Z-A) = -\tfrac{1}{2}\,(A-C)\times(Z-A) = -\tfrac{1}{2}\,\beta_Z\,\delta < 0,$$

where $(A-C)\times(Z-A)=(A-C)\times(Z-C)+(A-C)\times(C-A)=\beta_Z\,\delta$. So $\measuredangle(NC\to NZ)=-\angle CNZ$. $\square$

Both K and L are on the C-side of AB and on the B-side of AC (interior points; (2.2)). Hence:

$$\measuredangle(BA\to BK)=-\angle ABK=-\theta,\qquad \measuredangle(BA\to BL)=-\angle ABL,\qquad (3.1)$$
$$\measuredangle(CA\to CL)=+\angle ACL=+\theta,\qquad \measuredangle(CA\to CK)=+\angle ACK,\qquad (3.2)$$
$$\measuredangle(MB\to MK)=+\angle BMK=+\psi,\qquad (3.3)$$
$$\measuredangle(NC\to NL)=-\angle CNL=-\angle LNC=-\varphi.\qquad (3.4)$$

Use of "K inside angle LBA". Since L is interior to ABC, L is inside angle ABC (same side of AB as C and of BC as A, by (2.2)); by Lemma 5 applied to angle ABC ($0^\circ<\angle ABC<180^\circ$ as A,B,C form a triangle), $0^\circ<\angle LBA<\angle ABC<180^\circ$. Applying Lemma 5 to angle LBA and the point K inside it:

$$\angle ABL=\angle ABK+\angle KBL=\theta+\varphi',\qquad \varphi':=\angle LBK.$$

By hypothesis $\varphi'=\angle LNC=\varphi$. Hence, using (0.1) and (3.1),

$$\measuredangle(BK\to BL)\equiv\measuredangle(BA\to BL)-\measuredangle(BA\to BK)=-(\theta+\varphi)+\theta=-\varphi\pmod{360^\circ}.\qquad (3.5)$$

Use of "L inside angle ACK". K is interior to ABC, hence inside angle ACB, so $0^\circ<\angle ACK<\angle ACB<180^\circ$ by Lemma 5. Applying Lemma 5 to angle ACK and the point L inside it: $\angle ACK=\angle ACL+\angle LCK=\theta+\psi'$ with $\psi':=\angle LCK=\angle BMK=\psi$ by hypothesis. Hence by (3.2),

$$\measuredangle(CL\to CK)\equiv\measuredangle(CA\to CK)-\measuredangle(CA\to CL)=(\theta+\psi)-\theta=+\psi\pmod{360^\circ}.\qquad (3.6)$$

Two consequences ("nondegeneracy"): since $0<\varphi<180^\circ$ and $0<\psi<180^\circ$ by (2.3),

$$L\notin\text{line }BK\quad(\text{by }(3.5)),\qquad K\notin\text{line }CL\quad(\text{by }(3.6)),\qquad (3.7)$$

because if L lay on line BK (note $L\neq B$: L is interior to ABC), the ray BL would coincide with ray BK or its opposite, forcing $\measuredangle(BK\to BL)\in\{0^\circ,180^\circ\}$, contradicting (3.5) since $-\varphi\not\equiv 0^\circ,180^\circ\pmod{360^\circ}$; likewise for K and line CL (note $K\neq C$), using (3.6).

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4. The three circles

Recall line BY = line BK and line CX = line CL.

Claim 4.1 (ω_1). *The points B,C,X,Y lie on one circle ω_1.*

Proof. The points X,Y,B are pairwise distinct and non-collinear: if B were on line XY then (as $X\neq B$) line XY = line $XB=AB\ni Y$, contradicting $Y\notin AB$ (indeed Y lies on segment AC with $Y\neq A$, and $AB\cap AC=\{A\}$ because A,B,C are not collinear). Also $C\notin\{X,Y\}$. Now compute directed line angles mod 180°:

$\angle(BX,\,BY)\equiv\measuredangle(BA\to BK)\equiv-\theta\qquad$ (line $BX=AB$ since $X\in AB$, $X\neq B$; line BY = line BK),

$\angle(CX,\,CY) \equiv -\angle(CY,\,CX) \equiv -\measuredangle(CA \to CL) \equiv -\theta$ (line $CY = AC$ since $Y \in AC$, $Y \neq C$; line $CX =$ line

So $\angle(CX, CY) \equiv \angle(BX, BY) \pmod{180^\circ}$. By Lemma 2 (with $P = X,\ Q = Y,\ R = B,\ S = C$), C lies on the circle ω_1 through X, Y, B. $\square$

Since $X, B \in \omega_1$ are distinct points of line AB, and $Y, C \in \omega_1$ are distinct points of line AC, Lemma 1(i),(iii) give

$$\mathrm{pow}(A, \omega_1) = x\,c = y\,b. \qquad (4.1)$$

(Signed products: both X and B lie on the ray from A towards B, etc.) Also

$$\mathrm{pow}(B, \omega_1) = 0, \qquad \mathrm{pow}(C, \omega_1) = 0, \qquad (4.2)$$

and, since $B \neq Y$ are two points of ω_1 on line BK and $C \neq X$ are two points of ω_1 on line CL,

$$\mathrm{pow}(K, \omega_1) = \langle B - K,\ Y - K\rangle, \qquad \mathrm{pow}(L, \omega_1) = \langle C - L,\ X - L\rangle. \qquad (4.3)$$

Claim 4.2 (ω_K). *Let ω_K be the circumcircle of C, M, K (these are non-collinear: $K \notin$ line CM and $K \notin AB \supseteq \{M\}$ by (P1)). Then line AB meets ω_K exactly in the multiset $\{M, X\}$: if $X \neq M$ then $\omega_K \cap AB = \{M, X\}$; if $X = M$ then AB is tangent to ω_K at M.*

Proof. By (3.3) and (3.6), mod 180°:

$$\angle(CL,\,CK) \equiv \psi \equiv \angle(MB,\,MK) \ = \ \angle(AB,\,MK). \qquad (4.4)$$

Case $X \neq M$. The points K, X, C are pairwise distinct ($K \notin AB \ni X$; $C \notin AB$; $K \neq C$) and non-collinear: line $XC =$ line CL and $K \notin$ line CL by (3.7). Also $M \notin \{K, X\}$ ($M \in AB$, $K \notin AB$; $X \neq M$ by assumption). Using line $CX =$ line CL and line $MX = AB =$ line MB, (4.4) reads

$$\angle(CK,\,CX) \equiv -\psi \equiv \angle(MK,\,MX) \pmod{180^\circ}.$$

By Lemma 2 (with $P = K,\ Q = X,\ R = C,\ S = M$), M lies on the circle through K, X, C; that circle passes through the non-collinear points C, M, K, hence equals ω_K; thus $X \in \omega_K$. Since $M \neq X$ are two points of ω_K on AB, Lemma 1(i) gives $\omega_K \cap AB = \{M, X\}$.

Case $X = M$. Then $M \in$ line CL, so (as $C \neq M$) line $CL =$ line CM, and (4.4) becomes $\angle(AB,\,MK) \equiv \angle(CM,\,CK) \pmod{180^\circ}$. By Lemma 3(ii) (applied to $\omega_K \ni M, K, C$ and $\ell = AB$), AB is tangent to ω_K at M. $\square$

By Lemma 1(iii) (line AB through A and through B, meeting ω_K at M and X, coincident if $X = M$):

$$\mathrm{pow}(A, \omega_K) = \tfrac{c}{2}\,x, \qquad \mathrm{pow}(B, \omega_K) = (\tfrac{c}{2} - c)(x - c) = \tfrac{c}{2}\,(c - x), \qquad (4.5)$$

$$\mathrm{pow}(K, \omega_K) = 0, \qquad (4.6)$$

and, since $C \neq X$ are two points of ω_K on line CL,

$$\mathrm{pow}(L, \omega_K) = \langle C - L,\ X - L\rangle. \qquad (4.7)$$

Claim 4.3 (ω_L). *Let ω_L be the circumcircle of B, N, L (non-collinear: $L \notin$ line BN, $L \notin AC \supseteq \{N\}$ by (P2)). Then line AC meets ω_L exactly in the multiset $\{N, Y\}$ (tangency at N if $Y = N$).*

Proof. By (3.4) and (3.5), mod 180°:

$$\angle(BK,\, BL) \equiv -\varphi \equiv \angle(NC,\, NL) = \angle(AC,\, NL). \qquad (4.8)$$

Case $Y \neq N$. The points L, Y, B are pairwise distinct and non-collinear (line BY = line BK and $L \notin$ line BK by (3.7); $L \notin AC \ni Y$). Also $N \notin \{L, Y, B\}$. Using line BY = line BK and line $NY = AC$ = line NC, (4.8) reads

$$\angle(BL,\, BY) \equiv \varphi \equiv \angle(NL,\, NY) \pmod{180^\circ}.$$

By Lemma 2 (with $P = L,\ Q = Y,\ R = B,\ S = N$), N lies on the circle through L, Y, B, which then equals ω_L; so $Y \in \omega_L$ and $\omega_L \cap AC = \{N, Y\}$.

Case $Y = N$. Then $N \in$ line BK, so line BK = line BN, and (4.8) becomes $\angle(AC,\, NL) \equiv \angle(BN,\, BL) \pmod{180^\circ}$. By Lemma 3(ii) (with the circle $\omega_L \ni N, L, B$ and $\ell = AC$), AC is tangent to ω_L at N. □

Consequently

$$\mathrm{pow}(A, \omega_L) = \tfrac{b}{2}\, y, \qquad \mathrm{pow}(C, \omega_L) = \tfrac{b}{2}\, (b - y), \qquad \mathrm{pow}(L, \omega_L) = 0, \qquad \mathrm{pow}(B, \omega_L) = 0, \qquad (4.9)$$

and, since $B \neq Y$ are two points of ω_L on line BK,

$$\mathrm{pow}(K, \omega_L) = \langle B - K,\, Y - K \rangle. \qquad (4.10)$$

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5. The affine power identity

Define $F : \mathbb{R}^2 \to \mathbb{R}$ by

$$F(Z) := \mathrm{pow}(Z, \omega) - \mathrm{pow}(Z, \omega_K) - \mathrm{pow}(Z, \omega_L) + \mathrm{pow}(Z, \omega_1).$$

F is affine. For any circle Γ with centre Z_Γ and radius ρ_Γ, $\mathrm{pow}(Z, \Gamma) = |Z|^2 - 2\langle Z_\Gamma, Z\rangle + |Z_\Gamma|^2 - \rho_\Gamma^2$. In F the $|Z|^2$-terms cancel $(+1 - 1 - 1 + 1 = 0)$, so $F(Z) = \langle u, Z\rangle + v$ for a fixed vector u and scalar v.

$F(A) = 0$: by $\mathrm{pow}(A, \omega) = 0$, (4.5), (4.9), (4.1):

$$F(A) = 0 - \tfrac{c}{2} x - \tfrac{b}{2} y + cx = \tfrac{cx}{2} - \tfrac{by}{2} = 0,$$

using $cx = by$ from (4.1).

$F(K) = 0$: $\mathrm{pow}(K, \omega) = 0$ $(K \in \omega)$, $\mathrm{pow}(K, \omega_K) = 0$, and by (4.3), (4.10) $\mathrm{pow}(K, \omega_1) = \mathrm{pow}(K, \omega_L) = \langle B - K, Y - K\rangle$; the last two cancel.

$F(L) = 0$: $\mathrm{pow}(L, \omega) = 0$, $\mathrm{pow}(L, \omega_L) = 0$, and by (4.3), (4.7) $\mathrm{pow}(L, \omega_1) = \mathrm{pow}(L, \omega_K) = \langle C - L, X - L\rangle$.

Hence $F \equiv 0$: A, K, L are not collinear, so $K - A$ and $L - A$ are linearly independent; from $F(A) = F(K) = F(L) = 0$ we get $\langle u, K - A\rangle = \langle u, L - A\rangle = 0$, so $u = 0$, and then $v = F(A) = 0$.

Evaluating F at B and C and using (4.2), (4.5), (4.9):

$$\mathrm{pow}(B, \omega) = \mathrm{pow}(B, \omega_K) + \mathrm{pow}(B, \omega_L) - \mathrm{pow}(B, \omega_1) = \tfrac{c}{2}(c - x), \qquad (5.1)$$

$$\mathrm{pow}(C, \omega) = \mathrm{pow}(C, \omega_K) + \mathrm{pow}(C, \omega_L) - \mathrm{pow}(C, \omega_1) = \tfrac{b}{2}(b - y). \qquad (5.2)$$

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6. Conclusion

O is the centre of ω and $R_\omega = |OA|$ its radius, so $|OB|^2 = \mathrm{pow}(B,\omega)+R_\omega^2$ and $|OC|^2 = \mathrm{pow}(C,\omega)+R_\omega^2$. By Lemma 4 applied to the midpoints M of AB and N of AC:

$$4\,OM^2 = 2\,OA^2 + 2\,OB^2 - c^2, \qquad 4\,ON^2 = 2\,OA^2 + 2\,OC^2 - b^2.$$

Subtracting and using (5.1), (5.2) and $cx = by$ from (4.1):

$$4\,(OM^2 - ON^2) = 2\,(OB^2 - OC^2) - c^2 + b^2 = 2\big[\,\mathrm{pow}(B,\omega) - \mathrm{pow}(C,\omega)\big] - c^2 + b^2$$

$$= c(c-x) - b(b-y) - c^2 + b^2 = -cx + by = 0.$$

Therefore $OM = ON$. ■

Remarks on hypothesis usage

- $\angle KBA = \angle ACL \to$ Claim 4.1; $\angle LCK = \angle BMK$ together with "L inside angle ACK" $\to$ Claim 4.2; $\angle LBK = \angle LNC$ together with "K inside angle LBA" $\to$ Claim 4.3.
- "K inside triangle BMC" is used exactly for: K interior to ABC (sign facts (S1)–(S2), crossbar (P3)) and $K \notin$ line CM (ω_K nondegenerate). Similarly for L and line BN.
- That M, N are midpoints enters only through $AM = MB = \frac{c}{2}$, $AN = NC = \frac{b}{2}$ in (4.5), (4.9) and through Lemma 4.
- Numerical verification of every claim (generic and both tangency cases): `code/verify.py`, `code/check_signs.py`, `code/verify2.py`.